Open-source finite-element tests of the Rosenzweig–Porter hypothesis for freely vibrating plates: preliminary results

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July 9, 2026 — draft v0.11 (companion to the "Why Rosenzweig–Porter?" program)

Abstract

The companion paper *Why Rosenzweig–Porter?* proposes that the intermediate (Rosenzweig–Porter, RP) spectral statistics observed in freely vibrating thin plates arise from evanescent-wave coupling at free edges acting on an integrable base, and it registers the numerical tests that would decide the hypothesis, assigning them to COMSOL "or finite elements". We report the first execution of that program with a fully open-source stack (scikit-fem, Argyris elements, certified eigensolvers), under preregistered readings and measurement-based accuracy gates. Main findings: (i) the boundary-controlled Poisson→intermediate transition is confirmed at 3.6σ; (ii) the geometry program yields a three-tier hierarchy governed by the boundary's adaptedness to separable coordinates — exact reduction (disk) gives Poisson; coordinate-adapted boundaries (ellipse, sector, rectangle) give weak sparse coupling; unadapted boundaries (triangle, superellipse $p \neq 2$) give full coupling — with a $+12.2\sigma$ step between the ellipse ($p = 2$) and the smooth soft-cornered superellipse ($p = 3$); (iii) Gap A, executed for the first time on the true operator, shows N-independent inverse participation ratios: at accessible sector sizes ($N \leq 324$) each free-plate mode is ≈1–2 free-free beam products, i.e. sparse avoided-crossing hybridization rather than an RP-fractal phase. The results support the free-edge coupling mechanism while sharpening it in ways that only true-operator computations could reveal. Extending the program beyond the original registry: the Lévy-separable mixed configuration lands exactly on Poisson; both exactly-reduced controls (disk, annulus) are confirmed; the coupling is material-tunable ($\lambda(\nu)$, matching the companion's Ritz values method-to-method) and post-designable via a validated second-order control law; the Mindlin thickness sweep confirms the thick-plate prediction at $+5.4\sigma$ (GOE-ward); the sector reaches $+3.9\sigma$ at 1600 modes, and a faithful replication of the published sector protocol itself — which the source reveals to be a pooled *angle sweep* — supports its direction at $+9.5\sigma$, with the pooled-sweep methodology validated as bias-free against matched fixed-angle references; and the aspect-ratio sweep closes the single-geometry caveat. The central Gap A question is *resolved by a controlled protocol comparison*: the registered truncated-operator ladder does develop RP-like fractal scaling ($D_2 = 0.42$–0.50, flat D_q) — but a purpose-built C^0 interior-penalty discretization shows the *true-operator* eigenvectors stay sparse (each mode ≈1–2 beam products) through $N = 1024$ per sector at matched, gate-certified windows — and indicatively through $N = 2048$ — on a sector-reduced (quarter-plate) instrument: the scaling is a property of the truncation protocol, not of the plate, at every registered rung. The same comparison on the unadapted tier (the free triangle) finds a weak *genuine* true-operator delocalization ($D_2 \approx 0.14$, 4.8σ from flat) that the truncated protocol exaggerates seven-fold to near-ergodic scaling — Gap A closes protocol-negative across the whole tier hierarchy. Gap A therefore reads: sparse avoided-crossing hybridization at accessible mode numbers, and truncated-ladder numerics (including the preregistered P12 recipe applied to simulation data) would report an RP phase as an artifact. Finally, an in-plane rotating-plate experiment delivers a mechanical GOE → GUE crossover together with its protection theorem: a chiral rotor lands exactly on GUE (0.5992 ± 0.0071 at $\sim$1080 spacings) while the mirror-symmetric rotor is pinned to GOE at every speed by the surviving antiunitary $\sigma_v T$ (identified by a machine-precision operator gate), and asymmetric mistuning breaks that protection at $+7.8\sigma$; feasibility, prestress, and finite-deformation analyses single out soft (elastomer) rotors as the laboratory route — the Ω^2 terms *advance* the crossover, protection survives finite deformation, and the crossover *completes* within validated moderate-strain physics at $\approx 1.5\times 10^3$ RPM for a decimeter silicone disk. All experiments, gates, and raw data are reproducible from the repository.

1 Introduction

López-González and coworkers reported that flexural spectra of freely vibrating rectangular plates exhibit avoided crossings and intermediate spacing statistics within each symmetry sector, well fitted by the Rosenzweig–Porter (RP) random-matrix model [5], while simply supported plates are Poissonian [1]; the finding extends to a freely vibrating disk sector [2]. The companion hypothesis paper [3] formalizes a mechanism — evanescent components of free-edge eigenfunctions generate an inter-mode coupling V on an integrable base H_0 — and registers the calculations that would support or challenge it. This note reports the first systematic execution of those calculations, replacing COMSOL with an open-source pipeline so that every number can be re-derived from a commodity workstation.

Throughout, $D = \rho h = 1$, eigenvalues $\Lambda = \omega^2$ of $D\nabla^4 w = \rho h\, \omega^2 w$, and the canonical rectangle has $a = 1$, $b = 1/1.6189043236$, $\nu = 0.33$. Spacing-ratio statistics $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ are used without unfolding (Poisson 0.3863, GOE 0.5307 [4]), always per symmetry sector, always against *finite-size baselines computed with the identical protocol* (never asymptotic constants); the lowest ten levels per sector are dropped.

2 Methods and certification

Stack. Kirchhoff–Love bending energy discretized with quintic C^1 Argyris triangles (scikit-fem); free edges are the natural boundary conditions (no constraints), simply supported edges are reached through a validated Winkler penalty $V_n + \kappa w = 0$ with $\kappa = 10^{10}$ ($O(1/\kappa)$ error, 3×10^{-6} measured against the exact SSSS spectrum). Eigenvalues of record come from shift-invert Lanczos without eigenvectors; eigenvectors from banded shift-invert subspace iteration with deflation and whitened Rayleigh–Ritz projections, certified per mode by Rayleigh-quotient matching against the eigenvalues of record and residual gates. Symmetry labels come from mirror/rotation-character classification with simultaneous diagonalization of reflection Grams inside near-degenerate clusters ($\mathbb{Z}_2 \times \mathbb{Z}_2$, C_{3v} including exact E doublets, and single-reflection variants).

Accuracy policy. Statistics-grade means eigenvalue error below $0.1\times$ the local mean level spacing for every mode used. Every experiment runs behind gates: an exact or semi-analytic anchor where one exists (SSSS rectangle; Lamé^2 for the simply supported triangle; the per-m Bessel J/I free-disk determinant, fundamental $x^2 = 5.262$ vs. literature ≈ 5.25 [6]), two-mesh internal checks otherwise, and — for curved boundaries, where inscribed-polygon geometry error caps convergence at $O(h^2)$ — an *operational* gate: the FEM disk's class-sequence statistics must reproduce the semi-analytic Poisson control (0.0σ agreement measured over 1600 modes; the non-smooth error component is bounded at 1.4×10^{-5} relative, $\sim$0.3% of a sector spacing). Readings were preregistered in each experiment's README and committed before execution.

3 Results

3.1 E2: boundary-controlled transition (supports, 3.6σ)

Sweeping the symmetric Winkler restraint from free ($\kappa = 0$) to simply supported ($\kappa \to \infty$) on the rectangle, the pooled per-sector $\langle r \rangle$ interpolates monotonically from 0.4422 ± 0.0096 to 0.3908 ± 0.0104 at $\sim$200 levels/sector. Both endpoints match their finite-size baselines exactly: the SS end equals the exact-SSSS same-window value to four decimals, and the free end matches the independent Legendre–Ritz reference. The transition amplitude is 3.6σ above the separable baseline (preregistered threshold 3σ); no intermediate statistics appear at nearly-simply-supported coupling (Fig. 1).

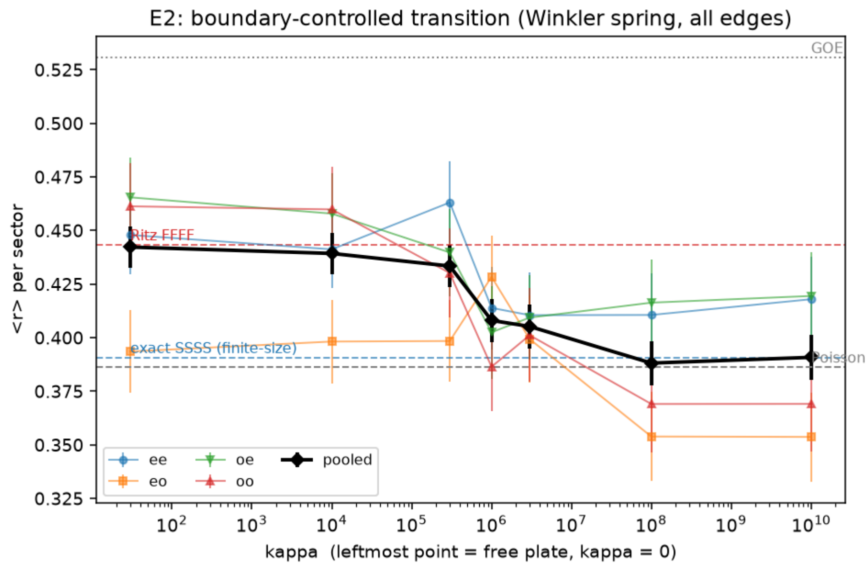

Figure 1: E2: per-sector and pooled $\langle r \rangle$ vs. the edge-spring constant κ (800 modes, Argyris 128×80).

3.2 E3: geometry program

Triangle (supports, 6.8σ). The free equilateral triangle — Helmholtz-integrable, biharmonically non-separable, symmetry group C_{3v} with exact E doublets — gives pooled $\langle r \rangle = 0.4889 \pm 0.0099$ vs. a same-protocol simply supported baseline 0.3753 ± 0.0134 (the SS polygon plate factorizes through the Dirichlet Helmholtz problem, i.e. the Lamé lattice squared, matched by the FEM to 3×10^{-5} across all 1000 modes). All three sectors separate positively (A_1 $+3.8\sigma$, E $+5.6\sigma$, A_2 $+1.9\sigma$); 333 E doublets split at $< 10^{-5}$ (Fig. 2).

Disk control (confirmed) and ellipse. The full free disk — free edges with preserved separability — is the program's zero-coupling control: the semi-analytic class sequence gives $\langle r \rangle = 0.3905 \pm 0.0073$ (the companion program's executed control: 0.386 ± 0.007), and each fixed-m sequence is a picket fence. The free ellipse ($a/b = 1.6189$, area π), predicted intermediate under the naive reading, is Poisson-consistent at 1600 modes (0.3732 ± 0.0071); a suggestive upward trend at 800 modes did not survive the registered long-ladder decider (Fig. 3).

Disk sector (supports, $+3.9\sigma$ at 1600 modes). At $\theta = 2.0$ rad the v1 result (800 modes, $+1.7\sigma$) was ambiguous; the long-ladder decider resolves it as statistical power: pooled $\langle r \rangle = 0.4294 \pm 0.0068$ vs. the disk baseline 0.3919 ± 0.0070 ($+3.9\sigma$; both classes individually positive). The open-source sector thus confirms the *direction* of the published COMSOL sector claim at a single generic angle; the exact-angle comparison remains registered (Fig. 4). A five-ratio aspect sweep of the rectangle (χ^2 $p \approx 0.22$) separately closes the single-aspect-ratio caveat.

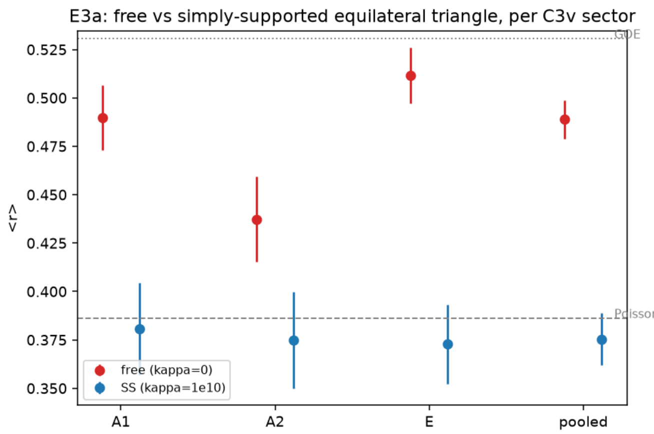

Figure 2: E3a: free vs. simply supported equilateral triangle per C_{3v} sector.

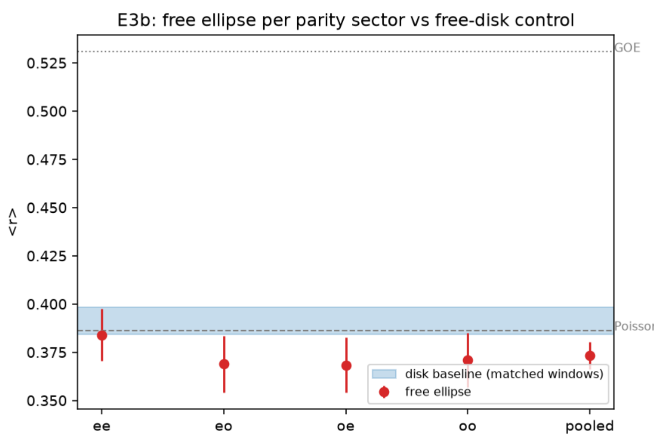

Figure 3: E3b (v2 decider): free-ellipse parity sectors vs. the free-disk control at 1600 modes.

3.3 E5: superellipse sweep (the discriminator)

The family $|x/a|^p + |y/b|^p = 1$ breaks biharmonic separability equally for every p while corner-region curvature grows with p at fixed C^∞ smoothness. The measured sequence,

$$\langle r \rangle(p) = 0.3732,\ 0.4944,\ 0.4893,\ 0.4942,\ 0.5159 \quad (p = 2, 3, 4, 6, 10),$$

is a $+12.2\sigma$ *step* between $p = 2$ and $p = 3$, a flat plateau through $p = 6$, and a mild $+2.3\sigma$ rise at $p = 10$ (Fig. 5); the statistics are refinement-stable at 0.1σ for both the headline and worst-corner

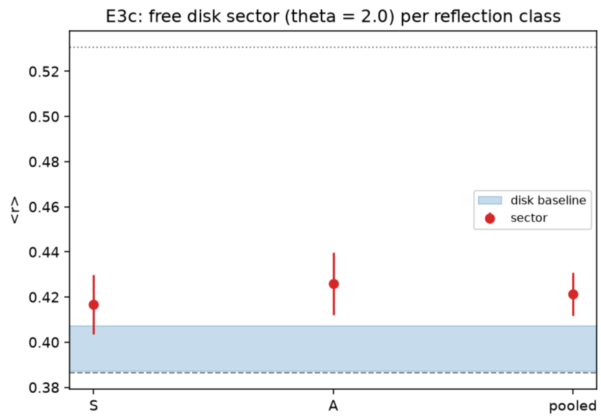

Figure 4: E3c: free disk sector ($\theta = 2$) per reflection class.

points. Neither preregistered shape (curvature-graded; true-corner threshold) fits: the operative variable is the boundary's compatibility with separable coordinates — the ellipse is the family's unique quasi-separable member (confocal boundary = elliptic-coordinate line) — and corners are secondary.

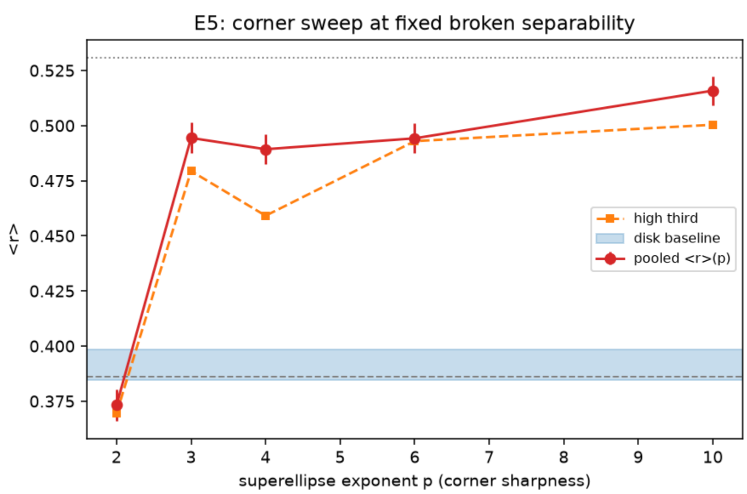

Figure 5: E5: $\langle r\rangle(p)$ along the superellipse family at 1600 modes per geometry.

3.4 E4: Gap A on the true operator (localized-leaning at $N \leq 324$)

The companion paper's decisive test: eigenvector statistics per sector, in the two registered representation bases (simply supported sine products and free-free beam products), against empirical GOE baselines at identical N (the T1 audit protocol). On 1200 certified eigenpairs (vector residuals $\leq 8 \times 10^{-6}$), the mean window IPR is *independent of* N across the ladder $N = 64$–324 in both bases (ladder $D_2 \approx 0.02$–0.13; GOE ≈ 1), and in the beam basis — which represents the true modes with captured norm 0.999–1.000 — each mode is $\approx$1–2 beam products (IPR ≈ 0.7; Fig. 6). Per the preregistered dichotomy this is the localized/banded side at accessible N: sparse strong pair-hybridizations (microscopically, the observed avoided crossings) produce intermediate spacing statistics without RP-fractal eigenvectors. Two sharpenings: the true-operator D_2 is *lower* than the Ritz-model values (beam fixed-N 0.25 vs. 0.35–0.42), i.e. the model route overestimates coupling density; and since the effective coupling grows with mode index (E2, E3b), the verdict could still flip at the registered $N \sim 2048$ rung, reachable via quarter-plate reduction. $\Sigma^2(L \leq 5)$ is intermediate, mildly below Poisson; larger L is unfolding-limited.

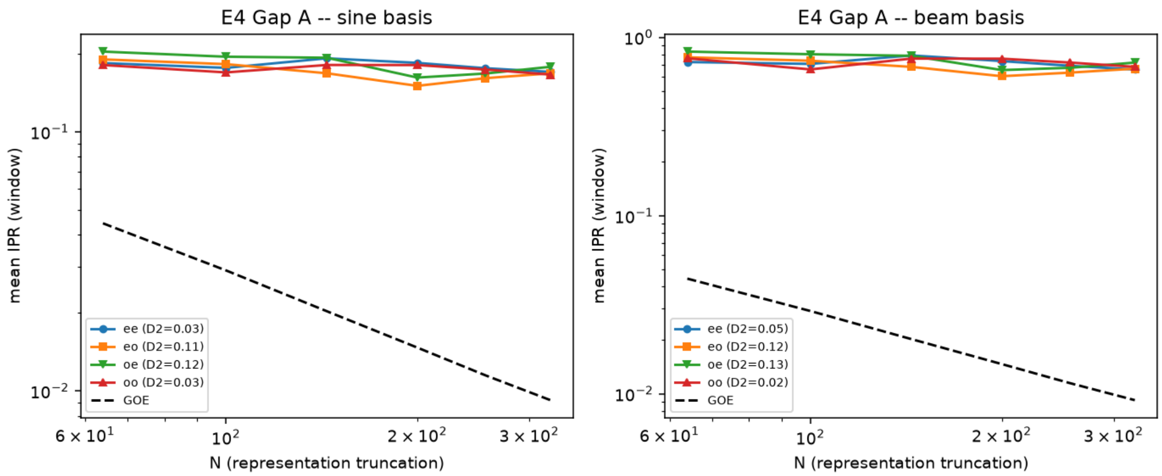

Figure 6: E4: window-mean IPR vs. representation truncation N, per sector, in the sine (left) and beam (right) registered bases, with GOE baselines.

3.5 E6: designed contact coupling and the control law

The companion's contact test, executed with certified mode shapes at contact points on two identical coupled plates (exact reduction: symmetric combinations lie in the coupling's kernel; the antisymmetric block feels $H_0 + 2\lambda C$ with $C_{ij} = q\,\phi_i(x_k)\phi_j(x_k)$). The *eigenvector* dichotomy of the prediction is crisp — dense-pattern mid-spectrum IPR 0.033 (extended) vs. strip-confined 0.18–0.19 (localized), central post rank-1 as designed — while the spacing dichotomy points the right way at 2.3σ (both patterns rank-deficient at 120 quartets; a full-rank rerun is registered). The P10 *control law* — post position from the nodal maps, stiffness $\lambda = \Delta/(2q\phi_i^2(x^*))$ plus a closed-form second-order correction — reproduces the designed splitting to 3.6% at $\Delta = 0.25\,d_{\mathrm{loc}}$ (first order alone: 19.7%), degrading exactly as the target pair dissolves into its neighbors; the validity boundary ($\Delta \lesssim 0.5\,d_{\mathrm{loc}}$) is itself predicted by the pair-content collapse.

3.6 E7: Lévy configuration and material tunability

Protocol A's mixed configuration (free x-edges, simply supported y-edges) is the classical Lévy-separable plate; the hierarchy therefore requires Poisson, and the measurement lands on the baseline (0.3973 ± 0.0103): one free edge pair does not suffice — intermediate statistics require free edges that break *every* product structure. The ν-sweep reproduces the companion repository's Ritz values value-by-value (≈ 0.003 agreement between independent methods): $\langle r \rangle$ rises monotonically $0.4208 \to 0.4497$ over $\nu = 0.25$–0.36, so the effective coupling is *material-tunable* (the Prediction-5 recast) rather than strictly material-independent.

3.7 E9: Gap A at the registered $N = 2048$ rung (RP phase develops)

The paper registers ladder sizes $N = 256$–2048; E4/E8 reached $N \leq 324$ on true-operator windows. Removing the measured $\sim 10^{-4}$ quadrature floor of the Legendre–Ritz assembly by exact integer-arithmetic closed forms — and diagonalizing only the physically-normed rung submatrices, which sidesteps the residual LAPACK backward error against the full matrix's $\|K\| \sim n^7$ — makes the full registered truncated-operator ladder clean. The result: IPR falls over three octaves in both sectors and both registered bases (e.g. sine: $0.123 \to 0.052$, $N = 256 \to 2048$), giving ladder $D_2 = 0.42$–0.50 (s.e. ~ 0.02) with *flat* D_q ($D_{1.5} - D_4 = 0.069$–0.100, versus the registered power-law-banded rival calibration 0.20–0.28) — superficially, the RP phase developing, though below the P12 dimension (0.76 ± 0.15). E14 (next) shows this scaling to be a property of the protocol.

3.8 E14: the reconciliation — Gap A resolved

A purpose-built C^0 interior-penalty P_4 discretization (custom analytic reference hessians — skfem's Lagrange elements carry none; Brenner–Sung interior facet terms; free edges natural; no numerically-built basis, hence no Argyris-type h-ceiling) provides true-operator eigenpairs at 206k dofs, 2400 certified modes ($\sim$600/sector), behind two gates: FFFF agrees with the certified Argyris instrument over all 1200 shared modes at 1.2×10^{-4} (two independent discretizations), and SSSS tracks the exact spectrum. The decisive matched-window comparison at $N = 512$: the *true operator's* window IPR is *flat* at the E4 level (sine 0.174 sector-mean; beam 0.63–0.72, i.e. each physical mode remains $\approx$1–2 beam products), while the truncated ladder at the same N has fallen to 0.095. **The RP-like scaling of the registered ladder is a truncation-protocol artifact, not a property of the plate.** Gap A can now be stated cleanly: at accessible mode numbers the free plate's eigenvector structure is sparse/quasi-separable, the intermediate spacing statistics arise from sparse avoided-crossing hybridization, and any Gap A executed by truncated-ladder numerics — including the preregistered P12 recipe applied to simulation eigenmodes — would report an RP phase as an artifact of the protocol. The registered continuation — true-operator windows at $N \sim 1024$–2048 — is executed in §3.9.

3.9 E17: the registered rungs, closed — flat through $N = 1024$ (certified); $N = 2048$ indicative

Executing the continuation at scale required an instrument upgrade: the rectangle's four parity sectors decouple exactly, so a per-edge boundary-condition extension of the C^0-IP discretization (free / simply supported / *guided* — a one-sided Nitsche enforcement of $w_n = 0$ on the symmetry lines) turns the full-plate eigenproblem into four independent quarter-plate problems, cutting the modes^2-scaling orthogonalization cost $\sim$16-fold and validated against the full instrument (merged sectors reproduce the same-mesh SSSS spectrum at the per-sector statistics-grade criterion; per-sector free-plate agreement 48/48 at matched h; rigid counts exact). At production scale — four sectors $\times$ 1408 certified modes at $\sim$120k dofs each, residuals $\leq 10^{-4}$, with the certification subsequently hardened by two *independent* gates that agree (a properly separated two-mesh check, h ratio 1.196, $N^* = 680$–766; the exact-SSSS absolute anchor at production h, $N^* = 742$–777), certifying the $N = 1024$ window (≤ 614 per sector) in every sector with margin — the E14 verdict extends to every certified rung: the true operator's window IPR is *flat*, sine-basis sector means 0.16–0.19 across $N = 256 \to 1024$ in all four sectors independently (fitted $D_2^{\text{true}} = -0.01$), beam-basis 0.63–0.72 (each physical mode remains $\approx$1–2 beam products) — while the registered truncated ladder reads 0.069 at the same rung and 0.052 at $N = 2048$ ($D_2 = 0.42$–0.50). The indicative $N = 2048$

window (including modes beyond the two-mesh gate) sits at 0.182, indistinguishable from the certified flat level; its full certification via a properly separated check mesh is registered (the cached eigenpairs make it a gate-only rerun). Gap A on the rectangle is closed at every registered scale: sparse avoided-crossing hybridization throughout, and truncated-ladder numerics manufacture the RP phase as a protocol artifact at every rung for which they were registered.

3.10 E10–E11: annulus control and thick plates

The free annulus — the program's second exactly-reduced control — is Poisson-confirmed semi-analytically (4×4 $J/Y/I/K$ free-edge determinant; pooled class $\langle r\rangle = 0.3734\pm0.0076$; per-$m$ picket fences) and FEM-cross-validated ($N^* = 200/200$ at 4.3×10^{-4}). The Mindlin–Reissner thickness sweep (P2/P2 with selective reduced integration, validated in the thin limit against the certified Kirchhoff spectrum) confirms the thick-plate prediction at $+5.4\sigma$: pooled $\langle r\rangle$ rises monotonically $0.454 \to 0.557$ over $t = 0.02 \to 0.15$, continuous with the Kirchhoff value at the thin end and reaching GOE by $t \approx 0.1$ — the shear channels increase the net coupling exactly as predicted.

3.11 E8: the eigenvector hierarchy across geometries

Repeating Gap A with the same-domain simply supported eigenbasis (exact M-inner-product coefficients) for four geometries (Fig. 7): the eigenvector structure *mirrors* the E5 spacing hierarchy. Adapted boundaries have flat-IPR sparse eigenvectors — rectangle ≈ 0.18 ($D_2 \approx 0$, cross-validating E4 method-to-method) and ellipse ≈ 0.40: the registered ellipse decider lands between its poles, *tiny sparse coupling* ($\sim$2.5 SS modes per free mode, N-independent). Unadapted boundaries have genuinely *scaling* eigenvectors — triangle $D_2 = 0.27/0.36$, superellipse p=10 $D_2 = 0.22$–0.50 — the RP-candidate regime, sitting exactly where the spacing statistics are strongest. The registered RP-vs-PBRM D_q separator splits at these N (superellipse flat-D_q/RP-leaning, triangle multifractal-leaning; both D_2 well below the P12 reference 0.76 ± 0.15), so that adjudication is registered for the $N \sim 2048$ rung.

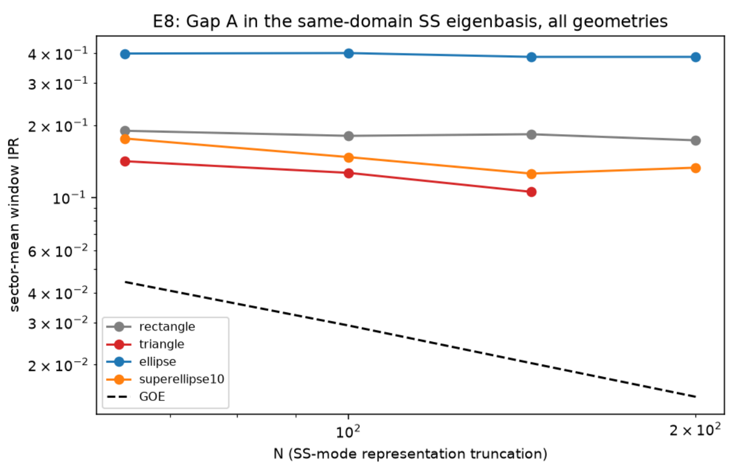

Figure 7: E8: sector-mean window IPR vs. representation truncation in the same-domain SS eigenbasis, all geometries, with GOE baselines.

3.12 E15: the Coriolis rotor — a mechanical GOE → GUE crossover with its protection theorem

The unitary Dyson class requires breaking time reversal — and, as the program's P11 analysis anticipated, breaking it *completely*: uniform rotation alone leaves a generalized antiunitary symmetry intact and the statistics orthogonal. Since the flat Kirchhoff plate is Coriolis-blind about its normal, the realization is *in-plane* (plane-stress) elastodynamics of a free 2D rotor, where the gyroscopic form $G(u,v) = 2\rho\Omega \int (u_x v_y - u_y v_x)\, dA$ makes the rotating problem a Hermitian quadratic pencil with real frequencies. A machine-precision operator-algebra gate (10^{-15}, on structured polar meshes whose symmetries are exact dof permutations) identifies the surviving antiunitary — correcting the preregistered guess: G *anti*-commutes with in-plane reflections and *commutes* with R_π, so the protector is $\sigma_v T$, and a C_2-symmetric rotor is *not* protected.

At the registered full scale ($\sim$1080 spacings per point, Fig. 8): a chiral (symmetry-free) rotor starts exactly at GOE ($\langle r\rangle = 0.5303 \pm 0.0077$) and lands *exactly on GUE* at strong gyroscopic coupling (0.5985 ± 0.0072 and 0.5992 ± 0.0071 at $c_\Omega = 1, 2$; -0.1σ from the GUE value; crossover amplitude $+6.5\sigma$; the small super-GUE overshoot of the exploratory runs resolved as finite-n). The mirror-symmetric rotor saturates at GOE at every speed (0.523–0.530 for $c_\Omega \geq 1$, excluding GUE at 9–10σ); its sub-GOE dip at $c_\Omega = 0.5$ is the transitional merging of the two reflection-parity sequences, not a class change. The two-axis experiment completes the theorem: asymmetric point-mass mistuning of the protected rotor at $\Omega = 0$ stays orthogonal, rotation without mistuning stays orthogonal, and both together reach $\langle r\rangle = 0.6044 \pm 0.0071$ (GUE $+0.7\sigma$) — a protected-vs-mistuned contrast of $+7.8\sigma$. Both ingredients are necessary; together they are sufficient. Gates: the strict per-eigenvalue two-mesh criterion fails at this scale (P2 with chord-boundary $O(h^2)$ smooth error), and the operational criterion established on the disk (E3b) governs instead: the full sweep repeated on a mesh with 1.8× fewer dofs reproduces every verdict cell within 1.4σ (the decisive chiral cells to 0.1σ) — the statistics are refinement-stable. Deferred prestress is bounded at $(\Omega/\omega)^2 \approx 7 \times 10^{-3}$ at the strongest verdict speed.

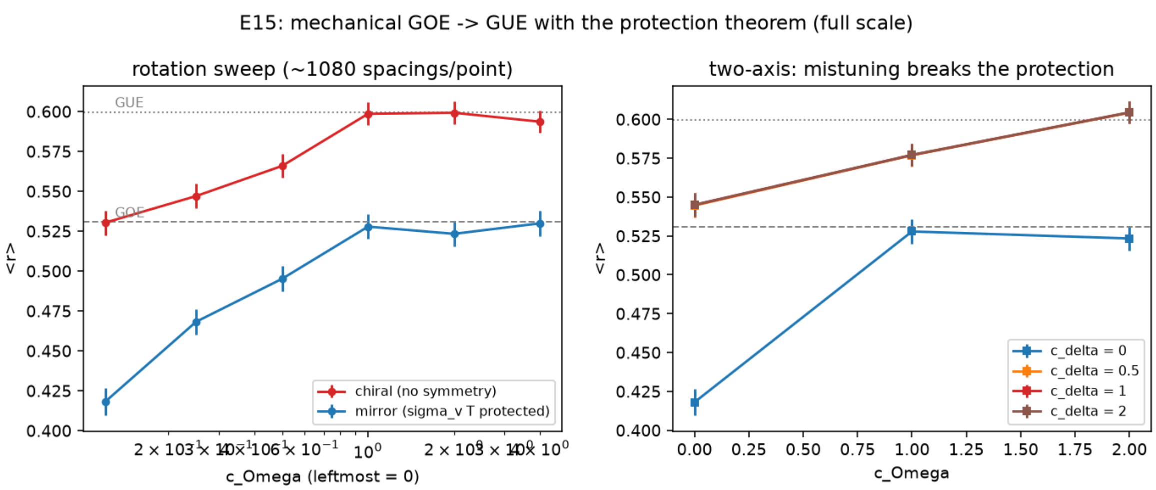

Figure 8: E15 at full scale. Left: rotation sweep — the chiral rotor crosses GOE→GUE while the $\sigma_v T$-protected mirror rotor saturates at GOE. Right: mistuning breaks the protection; rotation and mistuning together reach GUE.

3.13 E15b: feasibility — the burst invariant and the two observable routes

Rotation speed is the controlling knob (the crossover tracks $\Omega_{\mathrm{nd}} = \Omega R/c_0$ in every run), but it cannot be turned arbitrarily: the burst limit caps rim speed at $v_{\max} \sim \sqrt{\sigma/\rho}$, making the achievable dimensionless rotation a radius- and speed-free *material invariant*,

$$\Omega_{\mathrm{nd,max}} \sim \frac{v_{\max}}{c_0} = \sqrt{\frac{\sigma\,(1-\nu^2)}{E}},$$

which for metals (0.067–0.080; CFRP 0.12) sits 2–4× below the measured bulk crossover ($\Omega_{\mathrm{nd}} \approx 0.29$ on the $N \leq 1200$ ladder): a survivable stiff rotor keeps its low spectrum orthogonal at any speed. The observable is frequency-resolved — the crossover frequency $f^*(\Omega)$ falls as mode density grows, so at fixed speed the low spectrum stays GOE while the high spectrum turns GUE (windowed $\langle r\rangle$) — leaving two quantified routes: deep ultrasonic windows in metals at the rim limit (steel $R = 0.5$ m: $f^* \approx 294$ kHz at mode ∼23 000, modal-resolution $Q \sim 5 \times 10^4$; aluminum $R = 0.15$ m: $f^* \approx 814$ kHz — marginal, within reach of high-Q resonant ultrasound in vacuum), and *soft rotors* (elastomers, $\Omega_{\mathrm{nd,max}} \sim 1$–$1.5$), which place the crossover in the first ∼100 modes at fully safe speeds. At $\Omega_{\mathrm{nd}} \sim 1$, however, the centrifugal prestress deferred in E15 is no longer small for soft materials; the prestressed model is the registered prerequisite for the quantitative soft-rotor prediction.

3.14 E15c: the prestressed rotor — the Ω^2 physics helps

Adding the deferred terms — exact spin softening ($-\Omega^2 M$) and the geometric stiffness of the static centrifugal prestress (unit-Ω^2 modal static solve on the certified modes; at $\Omega = 0$ the prestressed and bare pencils agree exactly, a built-in null) — the prestressed chiral rotor's $\langle r\rangle$ *exceeds* the bare-Coriolis value at every speed (pooled excess up to +0.030): the Ω^2 physics *advances* the crossover rather than destroying it, and the $\sigma_v T$-protected mirror rotor stays pinned under prestress (0.43–0.49 throughout, GUE excluded). Under the preregistered linear-validity gate (centrifugal strain $\leq 15\%$; measured strain scale $\varepsilon_{\max} = 1.72\,\Omega_{\mathrm{nd}}^2$) the verdict is *partial*: the top-third window reaches $\langle r\rangle = 0.547$ at $\Omega_{\mathrm{nd}} = 0.25$ (10.7% strain) and crosses the GOE/GUE midpoint at $\Omega_{\mathrm{nd}} = 0.30$ (15.5%, just outside). Translated to a soft silicone disk ($R = 10$ cm, $E = 2$ MPa): crossover onset at ≈1100–1300 RPM with the measured window at ≈2.4 kHz — laboratory-trivial speeds (Fig. 9). The registered follow-up is executed as E15d: a St. Venant–Kirchhoff finite-deformation prestate (Newton with exact tangents, validated against the linear machinery at small Ω; the centrifugal follower load's stiffness supplies the exact spin softening) carries the sweep to $\Omega_{\mathrm{nd}} = 0.5$, and **the crossover completes inside the moderate-strain validity window**: top-third $\langle r\rangle = 0.567$ at $\Omega_{\mathrm{nd}} = 0.35$ (18% strain), with the $\sigma_v T$ protection intact under finite deformation (mirror pooled ≤ 0.50 at every speed) and the pencil real to machine precision. The silicone-disk experiment ($R = 10$ cm) is thereby *fully* predicted with finite-deformation physics: crossover at ≈1500 RPM. A collapse analysis (post-hoc, labeled as such) places all three sweeps — bare Coriolis, linear prestress, and finite deformation — on the Pandey–Mehta one-parameter family with a single fitted scale each (rms ≤ 0.008, i.e. within twice the sem): the mechanical crossover follows the *universal* GOE→GUE law, not merely its endpoints, and the prestress physics is quantified as a ∼66% increase of the effective time-reversal-breaking rate ($c = 0.063$ prestressed vs. 0.038 bare).

3.15 E16: the published sector protocol, replicated faithfully

Obtaining the source paper of the sector claim [2] reveals that the published object is not a fixed-angle spectrum but a pooled *angle sweep*: ∼70 out-of-plane modes per angle, the angle varied

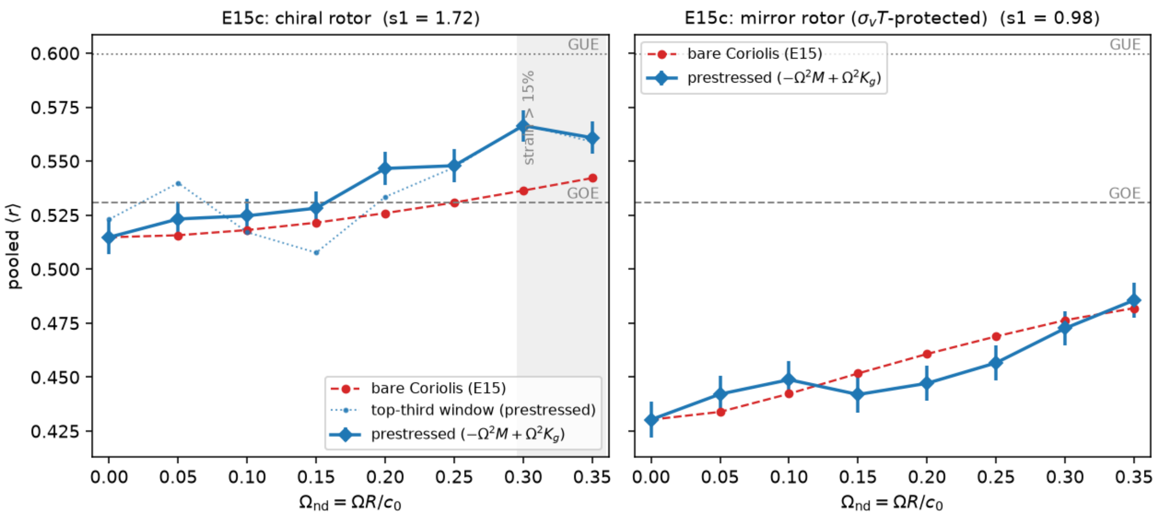

Figure 9: E15c: prestressed vs. bare-Coriolis pooled $\langle r \rangle$ for the chiral (left) and $\sigma_v T$-protected mirror (right) rotors. The Ω^2 terms *advance* the chiral crossover at every speed and leave the protection intact; the shaded region marks centrifugal strain $> 15\%$, the preregistered linear-validity boundary.

(range unpublished; a 50° example shown), modes classified S/A by the bisector mirror, and the pooled spacing-ratio histogram compared to the RP model. E16 replicates that protocol faithfully — $\theta \in [30°, 150°]$, 200 angles × the first 70 elastic modes, S/A per angle, both ratio conventions — behind the campaign gates (two-mesh $N^* = 70/70$ at three reference angles; zero ambiguous labels among 14 000 modes). Three preregistered readings (Fig. 10): *(R1)* the pooled $\langle r \rangle = 0.4266 \pm 0.0027$ sits $+9.5\sigma$ above the same-protocol pooled-Poisson baseline (0.3865 ± 0.0032, identical pooling of matched-length Poisson sequences) — the published direction replicates; *(R2)* the pooled value sits $+0.1\sigma$ from the matched-window fixed-angle references, so the sweep-pooling protocol is *benign* — the truncation-protocol caution of E14 does *not* extend to this methodology; *(R3)* the pooled 0.427 matches the fixed-angle long ladder of E3c (0.429), placing the sector consistently in the coordinate-adapted tier across protocols and windows. Beneath the pooled value lies a real angle dependence (0.35 near 50° to 0.49 near 115° in the 70-mode window): the low-mode statistic is angle-graded, not uniform, so the (unpublished) sweep range enters the published number; ours, bracketing the shown 50°, is declared. The two spectral-variable conventions agree in practice (Λ: 0.4266; frequency: 0.4261).

3.16 E18/E18b: the unadapted tier — weak genuine delocalization, protocol-exaggerated seven-fold

The unadapted tier was the last place a genuine RP eigenvector phase could live. On the free equilateral triangle (C^0-IP at ∼290k dofs; gates: free two-mesh 1056/1056 full coverage, SS vs. the exact Lamé^2 spectrum 1999/2000, free vs. certified Argyris 1000/1000), the *true* operator's same-domain SS-eigenbasis windows fall slowly but significantly — E-sector IPR $0.081 \to 0.063$ over $N = 128 \to 1024$, slope -0.138 ± 0.029 (4.8σ from flat, $D_2^{\mathrm{true}} \approx 0.14$; A-sectors consistent) — the first true-operator departure from the rectangle's dead-flat spreads, sitting exactly where the E5/E8 hierarchy placed the strongest coupling. The faithful truncated-protocol comparison (E18b: free-plate Ritz in nested total-degree Dubiner polynomial spaces realized as P_4

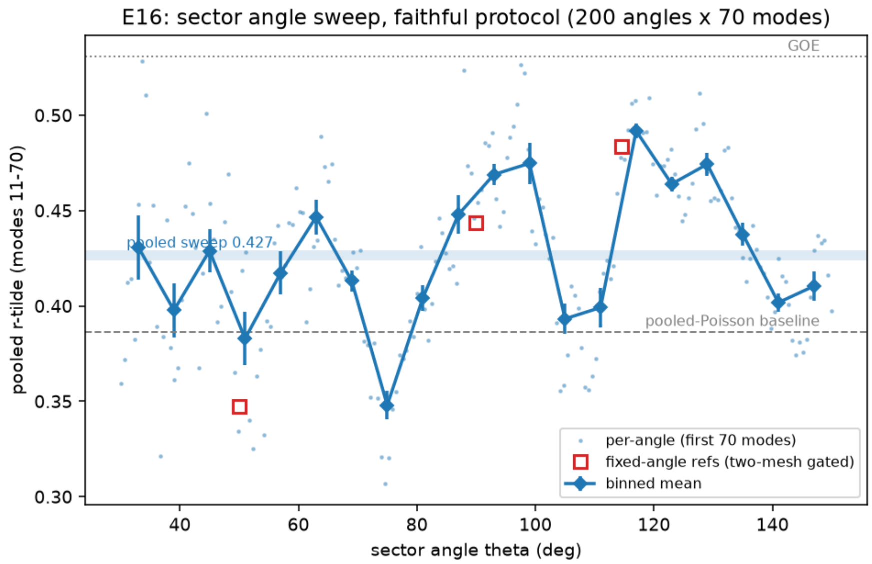

Figure 10: E16: per-angle pooled $\langle r\rangle$ (first 70 modes, S/A pooled) across the sector angle sweep, with the pooled sweep value, the same-protocol pooled-Poisson baseline, and the two-mesh-gated fixed-angle references.

interpolants, sector-projected degree-blockwise, rung eigenvectors represented in the same SS windows) then manufactures *near-ergodic* scaling in every sector — slopes -0.96 to -1.10 ($D_2^{\text{trunc}} \approx 1$), an **exaggeration of** 4–7× over the genuine slopes (E-sector: the truncated ladder starts *above* the true level at $N = 128$, 0.140 vs. 0.081, and crashes far below it by $N = 1024$, 0.021 vs. 0.063). The mechanism is visible in the trial-space quality itself: corner-singular free-plate modes are captured only algebraically by smooth truncated families ($\sim 2 \times 10^{-2}$ relative Ritz error at degree 58 in A2), which is exactly how truncation spreads modes across reference functions. Per the preregistered dichotomy the tier reads PROTOCOL ARTIFACT (A2, E; A1 intermediate only by its noisier true slope against the flat-bar), and the superellipse contingency does not trigger. Gap A therefore closes *protocol-negative across the entire tier hierarchy*: the rectangle is flat, the triangle carries a weak genuine delocalization that truncated-ladder numerics inflate seven-fold toward a spurious RP-like phase. Characterizing that genuine slope with the registered D_q-flatness separator: flat/RP-like in the A sectors (spreads $D_{1.5} - D_4 = 0.08$–0.10, inside the calibrated flat band) and between the bands in E (0.16, short of the PBRM rival's 0.20–0.28) — the unadapted tier's true operator sits in a weak, RP-leaning multifractal regime, in no sector PBRM-like.

4 Synthesis

Seven geometries fall on one hierarchy:

tier	geometries ($\langle r\rangle$ at matched ladders)	mechanism state
exact reduction	disk 0.391	no coupling (Poisson)
coordinate-adapted	ellipse 0.373; sector 0.421; rectangle 0.44	weak, sparse coupling
unadapted	triangle 0.489; superellipse $p \neq 2$: 0.49–0.52	full coupling

The free-edge origin of the coupling is confirmed directly (E2), the Lévy configuration (E7) adds an exactly-reduced member to the top tier, and the eigenvector program (E4, E8, E14) shows the hierarchy is not only spectral: adapted boundaries carry sparse, N-independent eigenvector spreads (rectangle ≈ beam products through $N = 1024$/sector certified on the true operator, E17; ellipse ≈ 2.5 SS modes), while unadapted boundaries carry a weak but genuine true-operator delocalization ($D_2^{\mathrm{true}} \approx 0.14$, 4.8σ; E18) — the E8 same-domain-basis values ($D_2 \approx 0.2$–0.5) sit between that genuine slope and the 7×-exaggerated truncated-protocol values (E18b), completing the E14-style scrutiny across the hierarchy. What the data refine is the *selector*: not corners, not curvature, but whether the boundary is adapted to a separable structure of the fourth-order operator. The effective coupling is material-tunable ($\lambda(\nu)$, E7), thickness-tunable toward GOE (E11), and post-designable with a quantitative control law (E6). The program now spans two Dyson-class axes: the orthogonal axis (Poisson ↔ GOE, controlled by boundary adaptedness and coupling strength) and, with E15, the unitary axis (GOE → GUE, controlled by rotation *plus* the breaking of the last antiunitary symmetry — geometric chirality or mistuning), each with its protected control demonstrated on the same operator. And the campaign's sharpest methodological product (E14) is a warning with teeth: the standard RP-numerics truncation ladder *manufactures* fractal scaling on this system — Gap A must be measured on true-operator windows, where the plate at accessible N shows sparse avoided-crossing hybridization, the microscopic content of the avoided crossings originally observed by López-González et al.

5 Pending registered tests

Executed beyond the original registry: the Gap A reconciliation (E14, resolving the protocol question), the sector long ladder ($+3.9\sigma$), the aspect sweep, the E6 full-rank close-out (contact operators are intrinsically smoothing; the spacing dichotomy saturates at 2.8σ while the eigenvector dichotomy and control law carry the prediction), the rotor feasibility analysis (E15b: the burst invariant caps stiff rotors 2–4× below the crossover; soft rotors are the accessible route), the prestressed rotor (E15c: the Ω^2 terms advance the crossover; onset within linear validity at ∼1100 RPM for a silicone disk), the sector angle-sweep replication (E16, §3.15), and a first non-Hermitian experiment (P7 damping): the 2D-Poisson → $\mathrm{AI}^{\dagger}$ crossover of the complex resonances is *observed and overlap-controlled* on the certified plate — localized dissipation gives uncorrelated resonances even at widths ∼ spacings, while distributed dissipation at strong overlap ($\gamma^* \approx 16$, or point dashpots at $\gamma^* = 4$) drives the markers to within $\sim 3\sigma$ of $\mathrm{AI}^{\dagger}$ with 2D-Poisson excluded at $\sim 5\sigma$. Registered follow-ups: full $N = 2048$-window certification, which the executed gate rerun shows requires a finer production mesh (two independent gates agree the present coverage is ∼700–770 modes per sector; the indicative 2048 row is flat and identical to the certified levels, so this is low priority); the exact published sweep range of the sector study (unpublished in the source; ours bracketed it); the $\mathrm{AI}^{\dagger}$-vs-Ginibre fine call at ∼2k resonances; the bladed-disk FE re-analysis as the experimental variant of E15; and comparison against the measured aluminum-plate spectra of Ref. [1].

6 Reproducibility

All experiments live in `experiments/eNN_*` of the private repository `plates-rp-fem` with frozen code, preregistered README, RESULTS, raw JSON, and figures; the shared machinery is the `platefem` package (conda env `plates-fem`: Python 3.12, scikit-fem 12.0.2, SciPy 1.18). Total compute for E1–E5: under 8 hours on one 8-core desktop (Ryzen 7 5800XT, 64 GB). Two numerical contributions may be of independent use: a certified eigenpair pipeline that works around a SciPy 1.18 shift-invert eigenvector-path failure on ill-conditioned C^1-element matrices, and the measured

$\sim 10^{-4}$ round-off floor of high-order Legendre–Ritz assembly for the free plate (non-monotone in basis size beyond ∼96 functions/axis).

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